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ICT-META: In-Context Aware Few-Shot Learner for Encrypted Traffic Classification

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ABSTRACT Encrypted traffic classification is a fundamental task in cybersecurity and network traffic management. With the popularity of deep learning, many studies have used it to achieve high accuracy in this field. However, these models rely on large amounts of labeled data and require retraining or fine-tuning when faced with new tasks, which is time-consuming and resource-intensive. Due to this limitation, few-shot traffic classification methods have emerged to reduce dependence on large labeled datasets by using only a few samples. However, these models often still require pre-training and fine-tuning to perform well. Meanwhile, Large Language Models (LLMs), such as ChatGPT, have attracted attention for their capacity to learn new concepts during inference without fine-tuning. Inspired by this, we propose In-Context Encrypted Traffic, a novel few-shot network traffic classification method allowing for few-shot classification without the need for fine-tuning. We use two existing encrypted traffic datasets to demonstrate the effectiveness of our method when facing new samples. We evaluate our method on a realistic 5-way few-shot classification task over encrypted traffic, and it achieves up to 5.7% higher accuracy than representative meta-learning baselines, without requiring fine-tuning.

INDEX TERMS Network security, encrypted network traffic, meta-learning, few-shot learning, in-context aware learning.

I. INTRODUCTION

ENCRYPTED network traffic flow classification is an important task in the field of cybersecurity, primarily focused on classifying encrypted network data for different types of network activities and applications. With the increasing variety of network services and the growing complexity of user behaviors, encrypted network data classification has become increasingly valuable and significant in society. It holds important applications in areas such as anomaly detection and intrusion detection [1], [2], illegal content monitoring [3], [4]user behavior analysis [5], [6] and traffic management and optimization [7], [8]. In many cases, traffic analysis attacks can be formulated as a classification problem, where the input to the classification model is a time series representing the statistical properties of traffic flows, such as packet size and direction. Previous research [9], [10], [11] has observed that deep learning models are particularly well-suited for this type of encrypted traffic classification task compared to traditional machine learning models.

However, deep learning-based traffic classification methods heavily rely on large amounts of labeled data, which is fundamentally at odds with the inherent difficulty of obtaining labeled data for encrypted traffic. In many real-world scenarios, the widespread adoption of privacy protection measures and encryption technologies has made labeling traffic samples both challenging and costly, exacerbating the issue of limited training data. Consequently, achieving efficient and accurate classification under the constraints of small-sample data has emerged as a core challenge in the field of encrypted network traffic classification. To address this issue, Few-Shot Learning demonstrates its unique advantages by efficiently modeling inter-class differences, allowing for rapid adaptation to new tasks. Remarkably, it can deliver strong performance even with limited data, making it a natural choice for tackling the small-sample problem in encrypted traffic classification.

The challenge of learning from very few training samples [12], [13] has led to the emergence of two main types of few-shot classification methods: 1) in-domain, and 2) cross-domain. In the in-domain setting, methods evaluate a meta-learner's ability to quickly adapt to new tasks after training on similar tasks. Models designed for this setting are typically fast but exhibit poor generalization to tasks outside the target domain [14]. Meanwhile, the cross-domain setting evaluates a meta-learner's ability to adapt to tasks in previously unseen domains. In this case, the model is trained on data from one category of network traffic, such as video or audio streams, and then evaluated on a different category of data, such as web traffic. Methods designed for this setting are highly adaptable but slower during inference as they require fine-tuning on the support set [15], [16]. Crucially, the meta-learners in both settings differ from humans in their ability to quickly generalize to new tasks. Similar to challenges in computer vision, within-domain meta-learning models perform well in encrypted network traffic classification in terms of adaptability, but cross-domain generalization still remains limited [14], [15], [17]. Additionally, existing few-shot learning methods for encrypted traffic classification also exhibit several critical limitations. First, most approaches rely on task-specific fine-tuning on the support set, which is impractical in real-world encrypted traffic scenarios where new applications and protocols emerge frequently [18], [19]. Second, prior few-shot ETC studies predominantly focus on in-domain evaluation, assuming similar traffic distributions between training and testing tasks, while cross-domain traffic classification involves substantial domain shifts that significantly degrade model performance [20], [21]. Third, existing methods adapt to new tasks primarily at the parameter level, rather than explicitly reasoning over the relationships between support examples and query traffic, limiting their ability to infer novel traffic patterns from limited context. In-context learning (ICL), initially popularized in natural language processing, enables models to perform few-shot classification by conditioning on a small set of labeled examples without parameter updates. Inspired by this, we propose to apply ICL to encrypted traffic classification, particularly focusing on cross-domain scenarios where the target traffic types were never seen during training.

In this paper, we address the challenges of few-shot and cross-domain classification in encrypted network traffic by proposing an In-Context Traffic Meta-learner (ICT-META). Our approach reformulates the n-way-k-shot few-shot traffic classification task as a non-causal sequence modeling problem over the support set and an unknown query data stream. Specifically, for an n-way classification task with k samples per class, we train a non-causal model on the support set consisting of traffic-label pairs, along with an unlabeled query traffic, to predict the label of the query traffic, allowing the model to learn new encrypted traffic features directly from the context during the inference stage, thus achieving the ability to classify new type of traffic without fine-tuning. Notably, in our experiments, the model was trained primarily on video and audio streaming traffic. To assess its cross-domain generalization ability, we conducted evaluations on a different type of traffic—encrypted web traffic—that was not seen during training. This approach leverages in-context learning to propose a new meta-learning approach, providing a new pathway for rapid and generalizable adaptation in the classification of encrypted network traffic. Accordingly, this work makes the following contributions:

- We propose the ICT-META model, a meta-learning approach with contextual awareness, to solve the problem of encrypted traffic classification in few-shot and cross-domain scenarios. Our model not only achieves strong performance on known tasks but also adapts effectively to new tasks without fine-tuning. To the best of our knowledge, this is the first approach that can perform cross-task encrypted traffic classification without requiring fine-tuning.
- We evaluate the effectiveness of our approach by conducting extensive experiments on a different type of traffic data than the data used during training. The results demonstrate that ICT-META outperforms existing methods, highlighting its potential for practical applications in encrypted network traffic classification.

The structure of the remaining chapters is as follows. In Section II, we introduce related works. Section III explains the structure of ICT-META in detail. Section IV describes the experimental setup. Section V presents the results of model evaluation and analysis. Finally, we conclude this paper in Section VI.

II. RELATED WORK

In this section, we review existing methods for the classification of encrypted network traffic and the application of meta-learning in this field, respectively.

A. ENCRYPTED TRAFFIC CLASSIFICATION

Encrypted traffic classification refers to the process of identifying and categorizing different types of network traffic without decrypting the data content, by analyzing statistical characteristics and temporal patterns of the traffic, so as to understand communication behaviors and support effective network resource management. Early studies mainly relied on shallow statistical features such as packet size and inter-arrival time for traffic classification [22], [23]. With the development of machine learning techniques, researchers have further incorporated traffic time-series features and protocol behavior characteristics to build classification models [17], [24], [25], thereby improving the recognition capability and classification accuracy for encrypted network traffic. The existing classification methods can be broadly divided into the following three categories based on their underlying modeling paradigms.

1) STATISTIC-BASED METHODS

Statistical-based methods employ predefined rules or probabilistic models to classify traffic based on feature distributions. Although early work on encrypted traffic fingerprinting demonstrates that such approaches can achieve good performance, they struggle with the complexity and variability of modern encrypted traffic [22], [23].

2) MACHINE LEARNING-BASED METHODS

Traditional machine learning-based methods, including Support Vector Machines and Random Forests, improve classification performance by learning non-linear decision boundaries from handcrafted features [24], [25]. Although these methods outperform purely statistical approaches, they remain highly dependent on feature engineering and typically require retraining to adapt to new applications or traffic patterns.

3) DEEP LEARNING-BASED METHODS

Deep learning-based approaches enable encrypted traffic classification by automatically learning hierarchical representations from raw or minimally processed traffic data. Early studies primarily adopted convolutional and recurrent neural networks to model packet-level and temporal patterns, as exemplified by DF [26], FS-Net [27], Deep Packet [28], TSCRNN [29], achieving improved accuracy over traditional machine learning methods but requiring large amounts of labeled data and task-specific retraining. With the introduction of transformer-based models, ET-BERT [17] applies self-supervised pre-training to learn contextualized traffic representations, significantly enhancing feature generalization and performance in low-label settings. More recent works further explore scalability and adaptability, such as MERLOT [30], which incorporates distilled large language models and mixture-of-experts for large-scale classification, and M3S-UPD [31], which leverages multi-stage self-supervised learning to support fine-grained classification and unknown pattern discovery. These models are particularly effective in scenarios with large-scale, well-labeled datasets. Yet, they face challenges due to their dependency on labeled data and limited adaptability to unseen traffic without costly retraining.

B. META-LEARNING

Meta-learning, or "learning to learn," aims to optimize models for rapid adaptation to new tasks with minimal data. Early frameworks, such as Neural Turing Machines [32] and self-attention mechanisms [33], laid the groundwork for sequence modeling and feature extraction. However, causal sequence modeling complicates feature optimization due to issues like permutation invariance [13].

Recent methods, such as model fine-tuning [34], [35], alleviate the scarcity of labeled data by leveraging pre-trained models, but they are resource-intensive and time-consuming, limiting their use in real-time systems. Recent meta-learning advances aim to improve task adaptation:

1) METRIC-BASED META-LEARNING

Methods like Prototypical Networks [36] compare new tasks with previously learned ones by optimizing the feature space for better generalization. However, they struggle with heterogeneous environments, such as encrypted network traffic, where data varies significantly.

2) OPTIMIZATION-BASED META-LEARNING

Methods like MAML [37] train models to adapt with minimal gradient updates. However, their reliance on fine-tuning makes them impractical for large-scale applications.

3) UNIVERSAL META-LEARNING FRAMEWORKS

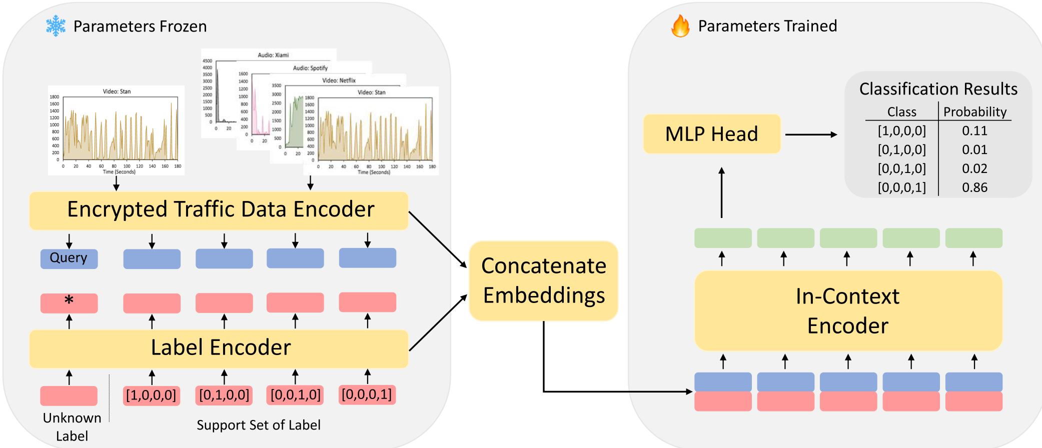
Frameworks [14] eliminate fine-tuning, enabling models to classify new tasks without task-specific training, aligning with context-aware meta-learning for encrypted traffic classification.

Despite their potential, existing methods for encrypted traffic classification face key limitations. Most rely on computationally intensive fine-tuning and large labeled datasets, hindering real-time application. They also lack cross-domain generalization and require retraining for even minor task changes. Thus, a fine-tuning-free method with strong cross-domain adaptability is urgently needed for dynamic, label-scarce environments.

C. IN-CONTEXT LEARNING

In-Context Learning (ICL) has recently emerged as a powerful paradigm that enables models to perform new tasks by conditioning on a small set of labeled examples provided at inference time, without updating model parameters. Initially popularized in NLP by large language models such as GPT-3 [38], ICL demonstrates that task adaptation can be achieved through contextual reasoning rather than explicit fine-tuning or retraining. However, the effectiveness of ICL depends on the model's prior knowledge, usually acquired from large-scale pre-training on diverse datasets, and on the size of the context window, which determines how many labeled examples can be considered simultaneously. Insufficient pre-training or an improperly sized context window may limit the model's ability to generalize to new tasks purely from in-context examples.

To improve few-shot generalization, MetaICL [39] meta-trains language models over diverse tasks to enable more effective in-context adaptation, while In-Context Tuning [40] frames task adaptation as a sequence modeling over input-label pairs. Theoretical analyses further show that transformer-based ICL implicitly implements task-specific learning algorithms, explaining its strong few-shot performance [41], and recent work explores unsupervised meta-learning via ICL, allowing models to acquire context-dependent representations from unlabeled data [42]. These



Class	Probability
[1,0,0,0]	0.11
[0,1,0,0]	0.01
[0,0,1,0]	0.02
[0,0,0,1]	0.86

FIGURE 1. ICT-META. ICT-META consists of three components: Encrypted traffic data encoder, Label encoder, and In-context encoder. The Encrypted traffic data encoder is used to extract feature representations of the support set and query set in encrypted network traffic data. The Label Encoder is responsible for encoding the labels of both the support set and query set. The extracted features and label embeddings are then concatenated and passed into the In-Context Encoder for processing. After going through multiple layers of self-attention and feature fusion in the Non-causal sequence model, the final output is fed into a shallow MLP for classification.

advances highlight the ability of transformers to generalize to new tasks purely through contextual reasoning, without parameter updates [43].

Inspired by these developments in NLP, recent works in encrypted traffic classification have explored context-aware and Transformer-based models to improve representation learning. For example, CoTNeT introduces a contextual Transformer network to capture inter-packet dependencies [44], while MIETT adopts a multi-instance Transformer architecture to aggregate packet-level representations [45]. Similarly, Language of Network models encrypted traffic as a structured "network language" and employs generative pre-training [46], whereas M3S-UPD leverages multi-stage self-supervised learning to efficiently capture fine-grained traffic patterns [31]. Although these approaches improve classification accuracy and reduce reliance on labeled data, they still require task-specific training or fine-tuning when applied to new traffic categories or domains.

Despite their empirical effectiveness, the aforementioned methods differ fundamentally from true in-context learning. They primarily enhance feature representations through contextual modeling, pre-training, or self-supervised learning, rather than enabling task adaptation purely at inference time. In contrast, ICL allows models to infer new classification rules from a small support set without any parameter updates, making it particularly suitable for few-shot and cross-domain scenarios. In encrypted traffic classification, where new applications and encryption protocols emerge frequently, large-scale data collection and labeling for every scenario is costly and impractical. By leveraging a small support set at inference time, ICL provides a more feasible solution for rapid adaptation to new traffic categories.

III. METHODOLOGY

Terminology: In the context of few-shot learning, the support set refers to a small number of labeled samples used to help the model perform classification, while the query set contains the unlabeled samples to be predicted. A frozen encoder is a feature extractor whose parameters remain fixed during training of the meta-learner and are not updated during task adaptation.

This section introduces our approach and outlines the key concepts of the proposed ICT-META method for encrypted traffic data classification. The main framework of ICT-META is shown in Fig 1, which consists of three primary components: the Encrypted Traffic Data Encoder, the Label Encoder, and the In-Context Encoder(i.e., a non-causal model, where we use a Transformer encoder).

A. PROBLEM FORMULATION

In few-shot encrypted traffic classification, we define each task $\mathcal{T}$ as an n-way k-shot classification problem, consisting of two subsets:

- A support set $\mathcal{S} = \{(x_i, y_i)\}_{i=1}^{n \cdot k}$, where $x_i \in \mathcal{X}$ is a traffic sample and $y_i \in \mathcal{Y} = \{1, 2, \ldots, n\}$ is its label;
- A query sample $x_q \in \mathcal{X}$, whose label $y_q \in \mathcal{Y}$ is unknown.

The goal is to predict y_q given $\mathcal{S} \cup \{x_q\}$, under the constraint that each class in $\mathcal{Y}$ has only k labeled samples in $\mathcal{S}$.

Formally, the classification function is defined as:

$$f(x_q; \mathcal{S}) = \arg\max_{y \in \mathcal{Y}} P(y \mid x_q, \mathcal{S})$$

We reformulate this few-shot task as a non-causal sequence modeling problem, where the model receives a concatenated sequence of support instances and a query instance:

$$\mathcal{X}_{seq} = [(x_1, y_1), \ldots, (x_{nk}, y_{nk}), (x_q, y_{unk})]$$

where y_{unk} is a learned "unknown" token.

The model is trained to predict the correct label for x_q by leveraging the contextual relationships between the query and the support examples.

B. IN-CONTEXT TRAFFIC META-LEARNER(ICT-META)

We propose ICT-META as a meta-learning framework designed to perform few-shot encrypted traffic classification without requiring fine-tuning. The core idea is to reformulate the typical n-way k-shot classification task as a non-causal sequence modeling problem that jointly considers both the support set and the query instance in a unified context.

It is worth noting that each traffic sample x_i is represented by a one-dimensional sequence of packet sizes as the feature representation. To standardize input length, sequences are zero-padded or truncated to 500 bytes. This choice is motivated by several factors. First, packet size falls into the category of metadata, which can still be obtained easily after encryption, making it one of the few directly observable key pieces of information. Moreover, compared with other statistical features such as byte size, packet size as a sequence feature achieves better classification results. Second, different types of encrypted traffic, including video, audio, and web pages, exhibit unique distribution patterns of packet sizes. For example, video streams generate packet sequences with "periodic size fluctuations" due to dynamic bitrate adjustment, audio streams may present a size pattern of "an initial large peak followed by stability" and web browsing features more random intervals and peaks in packet size sequences due to user interactions such as clicking and loading. These patterns can be directly characterized by one-dimensional packet size sequences, which serve as the core basis for distinguishing traffic types [4]. Finally, a one-dimensional packet size sequence can be naturally mapped to the temporal/vector input of the model, eliminating the need for complex high-dimensional feature transformation while balancing classification performance with computational cost.

For each task, ICT-META takes as input a support set of n×k encrypted traffic samples with their corresponding labels, and a query sample whose label is to be predicted. Each traffic sample is first passed through a frozen Encrypted Traffic Data Encoder (based on a residual network), which extracts fixed-dimensional feature vectors. In parallel, the corresponding class labels in the support set are embedded using a Label Encoder, while the query label is represented using a learnable "unknown" token.

The support features, label embeddings, and the query feature are then concatenated into a unified sequence and passed into the In-Context Encoder, implemented as a non-causal Transformer encoder. Unlike causal models that enforce strict input order, the non-causal design allows ICT-META to process support and query samples symmetrically and holistically. This is particularly suitable for encrypted traffic classification, where packet-level sequences can vary due to retransmissions, multiplexing, or client-side processing delays, making temporal order brittle and potentially misleading. By treating the support-query pair as an unordered set of embeddings, the model focuses on discriminative feature relationships rather than fragile sequence dependencies.

Finally, the output of the In-Context Encoder is passed through a lightweight MLP classification head, which predicts the class label of the query sample. Importantly, only the In-Context Encoder and MLP are trained; both the traffic encoder and label encoder remain frozen, allowing ICT-META to generalize across tasks without any parameter updates during inference.

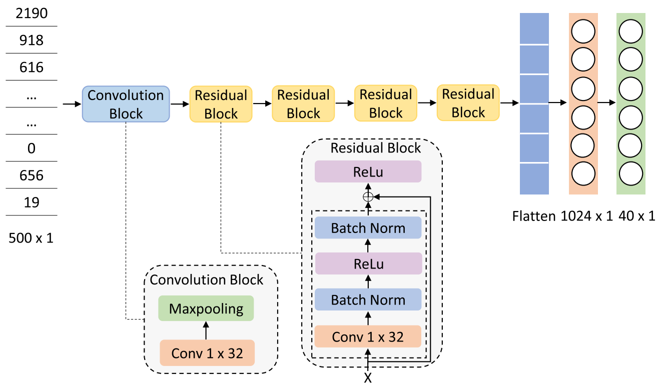


FIGURE 2. **Encrypted traffic data encoder. The Encrypted traffic data encoder is used to extract features from encrypted network traffic data.**

C. ENCRYPTED TRAFFIC DATA ENCODER

The Encrypted Traffic Data Encoder is built on a residual network (ResNet) architecture to tackle challenges in training deep neural networks. By introducing shortcut connections, ResNet mitigates vanishing gradients, enhances gradient flow, and improves training efficiency and model performance [47], as shown in Fig. 2. Its core component is the residual block, comprising two convolutional layers with batch normalization and ReLU activations. The shortcut connection directly adds input to the output, preserving information and enabling deeper feature learning.

The encoder begins with a convolutional layer for initial feature extraction, followed by multiple residual blocks that capture increasingly abstract features and long-range dependencies critical for understanding encrypted traffic behavior. Shortcut connections help retain vital information during deep training [48]. Finally, fully connected layers map the extracted features to output classes, supporting accurate classification.

In essence, this encoder leverages ResNet's strengths—shortcut connections and deep feature learning—to achieve robust performance in encrypted traffic classification.

D. LABEL ENCODER

1) LABEL ENCODER DESIGN AND EMBEDDING CONSTRUCTION

In few-shot encrypted traffic classification, each task contains only a limited number of labeled support samples, and encrypted traffic features often exhibit high intra-class variability and inter-class similarity. Under such conditions, relying solely on traffic features makes it difficult for the model to learn stable decision boundaries. Therefore, we introduce a Label Encoder to transform discrete class labels into continuous embeddings, providing an explicit geometric prior that guides the model's discriminative reasoning.

Label embeddings enable the In-Context Encoder to jointly model both traffic features and label information through self-attention. This allows the query sample to be aligned not only with support features but also with the corresponding class representations, enhancing classification robustness under data scarcity. Formally, the embedding matrix is denoted as $\mathbf{E} \in \mathbb{R}^{C \times p}$, where C is the number of classes and p is the embedding dimension. Accordingly, we organize the Label Encoder as a unified module and investigate two representative label embedding construction strategies, which are described as follows.

a) ELMES ENCODING

The ELMES embedding matrix is precomputed using the Equal Length Maximum Equiangular Set (ELMES) encoding mechanism [49], which aims to enforce equal vector lengths and maximize equiangular separation among class embeddings. This design is motivated by the need to address the fuzzy class boundaries and feature overlap in encrypted network traffic classification. MeanWhile, equal vector lengths eliminate norm disparity interference, ensuring the model's similarity judgment reflects true class differences rather than arbitrary scale variations. Maximal equiangularity minimizes embedding distribution overlap, reducing the model's uncertainty in class detection and enhancing discriminative power—critical for scenarios with limited labeled traffic data. Additionally, ELMES maintains permutation invariance of the input sequence, ensuring stable predictions regardless of the order of traffic samples in the input.

Specifically, a random orthogonal matrix $U \in \mathbb{R}^{p \times C}$ is first generated via QR decomposition. An adjustment matrix M^* is then constructed as

$$M^* = \sqrt{\frac{C}{C-1}} \cdot \left(I_C - \frac{1}{C}\mathbf{1}\mathbf{1}^\top \right), \tag{1}$$

where I_C denotes the identity matrix and $\mathbf{1} \in \mathbb{R}^C$ is an all-ones vector. The embedding matrix is finally computed as

$$\mathbf{E} = (U \cdot M^*)^\top. \tag{2}$$

The resulting embedding matrix remains fixed during training, ensuring equal length and maximal equiangularity among class vectors.

b) RANDOMLY INITIALIZED EMBEDDINGS

For comparison, we also employ a randomly initialized embedding matrix. This matrix is initialized using the He initialization scheme [50], which is defined as:

$$\mathbf{E}_{ij} \sim \mathcal{U}\left(-\sqrt{\frac{6}{\mathrm{fan\_in}}}, \sqrt{\frac{6}{\mathrm{fan\_in}}} \right), \tag{3}$$

where fan_in represents the number of input units in the embedding layer. In this case, fan_in = $C+p$, where C is the number of classes and p is the embedding dimension. This initialization ensures that signals propagate stably through the network during training. Unlike ELMES, the embedding matrix initialized in this manner is learnable and updated dynamically during the training process.

2) LABEL ENCODING PROCESS

Once the embedding matrix $\mathbf{E}$ is constructed, the label encoding process involves two steps:

a) ONE-HOT ENCODING

Discrete labels $\mathbf{y} \in \mathbb{R}^B$ are transformed into one-hot encoded representations:

$$\mathbf{Y} = \mathrm{OneHot}(\mathbf{y}) \in \mathbb{R}^{B \times C}, \tag{4}$$

where B is the batch size.

b) EMBEDDING TRANSFORMATION

The one-hot matrix $\mathbf{Y}$ is projected into the embedding space:

$$\mathbf{Z} = \mathbf{Y} \cdot (\alpha \cdot \mathbf{E}), \tag{5}$$

where α is a learnable scaling factor initialized to 1, and $\mathbf{Z} \in \mathbb{R}^{B \times p}$ is the resulting label embedding matrix.

3) COMPARISON OF ENCODING METHODS

We conducted comparative experiments to evaluate the performance of ELMES encoding and the randomly initialized embedding matrix. The detailed experimental setup and results are presented in the experimental design section of Section V.

E. IN-CONTEXT ENCODER

In-context learning leverages the relationships between a small support set and a query set to perform accurate inference without task-specific fine-tuning. To effectively implement this approach in encrypted network traffic classification, the In-Context Encoder is designed to fully utilize the global contextual information within the support and query sets. By simultaneously processing both sets, the encoder ensures that the model dynamically adapts its predictions based on contextual cues provided by the support set. This design enhances the model's robustness and generalization capabilities, which are crucial under few-shot conditions.

The In-Context Encoder is built upon the Transformer architecture and employs multi-head self-attention to capture dependencies across both support and query samples. Specifically, the encoder adopts a non-causal formulation at the task inference level, where autoregressive constraints are removed to allow bidirectional attention over the entire support–query context. This design enables the model to infer task-specific decision rules by jointly reasoning over labeled support samples and unlabeled queries. To respect the set-based nature of few-shot learning, the support set is treated as an unordered collection of examples. Accordingly, positional encodings are excluded for support samples, ensuring that model predictions are invariant to permutations of the support set. Importantly, this design does not discard structural information within individual traffic samples, as the temporal or feature order inside each encrypted traffic sequence is preserved and encoded by the backbone feature extractor. By disentangling task-level contextual reasoning from sample-level structural encoding, the proposed encoder flexibly models support–query relationships without introducing ordering bias.

In the specific implementation, the In-Context Encoder is composed of multiple layers of Transformer encoders. The concatenated feature sequence is fed into the In-Context Encoder for processing. Each encoder layer consists of two main components: Self-Attention and a Feedforward Neural Network, which is implemented as a Multi-Layer Perceptron (MLP). In each layer, the input is first normalized using LayerNorm, followed by Self-Attention computation via the MultiheadAttention layer to capture dependencies between different positions in the input sequence. The output of the self-attention mechanism is then added to the input through a residual connection. Next, the input is processed by another LayerNorm and MLP layer to further enhance its feature representation capacity. The output of each layer is again added to the input through another residual connection, which helps mitigate the vanishing gradient problem during training. Finally, the output sequence from the Transformer encoders is standardized through the final LayerNorm, yielding the final encoded representation. The specific module architecture is shown in Fig. 3.

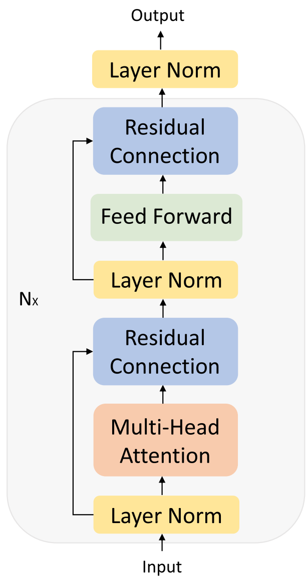


FIGURE 3. In-context encoder architecture.

IV. EXPERIMENTS

A. DATASETS

1) DC DATASET

The DC dataset [4] comprises two parts (9000 and 15000 samples), collected over WPA2-encrypted 2.4 GHz WiFi from video (Netflix, Stan, YouTube), audio (YouTube Music, Spotify, Xiami), and web (Wikipedia, ABC News, SMH) platforms. MAC-layer features such as frame size, type, and signal strength are used for analysis. Its diversity makes it suitable for evaluating cross-domain generalization across various encrypted traffic types.

2) GOOGLE HOME DATASET

The Google Home dataset [51] includes 100 voice command samples, aiding ICT-META training. Its rich contexts enhance the model's ability to generalize and respond to voice-based encrypted traffic in dynamic, real-world scenarios.

B. FEATURE ENGINEERING

For both datasets, we adopt the packet size as the only feature, following the approach introduced by Li et al. [4], who demonstrated that packet sizes can reveal class-specific patterns even in fully encrypted traffic. Each traffic sample is treated as a one-dimensional sequence of packet sizes (in bytes). This design choice avoids reliance on timing, direction, or protocol-level metadata, and ensures applicability under strict encryption scenarios.

To ensure consistent input across samples, all sequences are either zero-padded or truncated to a fixed length of 500 before being passed into the model. No additional features such as inter-arrival time, directionality, or signal strength are used. This unified feature representation allows us to directly compare model performance across multiple domains and datasets.

C. EXPERIMENTAL DESIGN

1) ENCRYPTED TRAFFIC DATA ENCODER

Our encrypted traffic data encoder uses a well-trained residual network. The training dataset includes multimedia data from the DC Dataset, with 30 video content items from YouTube Video and 10 audio content items from YouTube Music. This diverse dataset provides ample training samples, enabling the network to efficiently extract features from encrypted network traffic.

TABLE 1. Experimental settings for ICT-META.

Stage	Parameter
Training	Optimizer: Adam
	Learning rate: 0.0001
	Learning rate scheduler: Custom Cosine
	Batch size: 75
	Epochs: 40
Meta-Testing	Way: 5
	Shot: 5 or 1
	Query shot: 16
	Trials: 4,000

2) LABEL ENCODER
We compared the performance of ELMES encoding and a randomly initialized embedding matrix. Experimental results (Table 2, 1CT-META*) show that both methods perform similarly in some cases, but ELMES encoding demonstrates superior class separation due to its geometric properties, making it more effective for tasks requiring robust class discrimination.

3) ICT-META
To evaluate the cross-task adaptability of ICT-META, we deliberately constructed the dataset to ensure that the training and testing sets belong to different service categories (e.g., training on video/audio and testing on web traffic). To train the ICT-META model, we used two diverse datasets: DC Dataset and the Google Home Dataset. The DC Dataset includes video, audio, and web stream data, with video stream data sourced from Netflix and Stan, and audio stream data from Spotify and Xiami. The Google Home Dataset consists of 100 speech command samples. We used the video and audio stream data from DC Dataset, combined with the Google Home Dataset, for model training. Additionally, we leveraged DC Dataset's web stream data (from sources such as ABC, SMH, Wikipedia, and Web—where ABC, SMH, and Wikipedia were combined into a single dataset) to assess the model's ability to generalize classification tasks for unseen domain data without requiring fine-tuning. In summary, the ICT-META training dataset not only enriched the model's training data but also strengthened its ability to handle cross-domain classification tasks, enabling effective generalization without requiring fine-tuning.

4) ABLATION STUDY
To investigate the contribution of the Google Home Dataset in training, we also conduct an ablation experiment where ICT-META is trained only on the DC Dataset (video, audio stream data) without the Google Home Dataset. The model is then evaluated on web stream data to assess cross-domain performance.

D. EXPERIMENTAL ENVIRONMENT
The model training for this work was conducted on a local server with the following configuration: a 12th Gen Intel® Core™ i7-12700F processor (2.10GHz), 32GB RAM, Ubuntu 22.04 as the operating system, and an NVIDIA GA104GL [RTX A4000] GPU. The implementation of ICT-META is based on Python 3.7 and the PyTorch 1.13.1 deep learning framework.

E. COMPARED METHODS
Among the compared baseline methods, several were originally proposed in the computer vision domain, such as MetaNIW. However, these methods are not inherently image-specific; instead, they operate on generic feature embeddings and perform task-level inference over support–query sets. To adapt them to encrypted traffic classification in a fair and consistent manner, we remove their original image encoders and replace them with the same encrypted traffic feature encoder used in our ICT-META framework. As a result, all methods receive identical fixed-dimensional traffic representations as input, and the observed performance differences mainly reflect their task adaptation mechanisms rather than feature extraction advantages.

1) METANIW
MetaNIW (Meta Neural Information Worker) [52] is a meta-learning method for few-shot tasks that models task-level features to support rapid adaptation. It performs well in low-resource scenarios by generalizing effectively from limited samples.

2) SNAIL
SNAIL (Simple Neural Attentive Meta-Learner) [53] combines temporal convolutions and attention mechanisms for efficient few-shot learning. Temporal convolutions capture context, while attention highlights key features, enabling fast and effective adaptation.

3) GPICL
GPICL (Gaussian Process Incremental Contrastive Learning) [43] merges Gaussian processes with contrastive learning to handle incremental learning. It models uncertainty and learns robust feature representations, maintaining stable performance amid changing data distributions.

4) METAMRE
MetaMRE [19] is a few-shot cross-domain encrypted traffic classification framework based on multi-task representation learning. It enhances feature discriminability through supervised and auxiliary tasks and adopts a gradient-based meta-learning strategy for rapid adaptation to new traffic categories. During meta-testing, task-specific fine-tuning is performed on the support set. For fair comparison, we implement MetaMRE under the same 5-way K-shot cross-domain setting.

TABLE 2. Performance comparison of methods on 5-way *K*-shot datasets (Accuracy %).

Method	abc		wiki		smh		web (abc,wiki,smh)	
	5-way 5-shot	5-way 1-shot	5-way 5-shot	5-way 1-shot	5-way 5-shot	5-way 1-shot	5-way 5-shot	5-way 1-shot
MetaNIW	72.221±0.262	59.766±0.351	79.107±0.291	71.013±0.214	**63.536±0.324**	**54.008±0.364**	79.936±0.238	70.219±0.350
SNAIL	20.236±0.245	20.204±0.352	20.085±0.348	20.120±0.345	20.278±0.262	19.839±0.258	20.441±0.313	19.930±0.375
GPICL	35.368±0.307	38.053±0.344	47.146±0.401	61.824±0.393	36.254±0.267	41.746±0.311	47.773±0.409	62.408±0.413
MetaMRE†	73.322±0.527	60.268±0.519	80.965±0.503	72.094±0.522	**66.732±0.541**	**56.229±0.573**	85.093±0.501	70.941±0.562
ICT-META*	66.350±0.321	56.338±0.335	81.229±0.278	**74.448±0.334**	61.537±0.321	51.336±0.351	80.716±0.278	72.641±0.345
ICT-META	**73.913±0.312**	**63.782±0.368**	**81.565±0.257**	71.315±0.337	58.792±0.331	49.330±0.348	**85.668±0.250**	**75.895±0.348**
ICT-META* w/o	64.112±0.318	55.997±0.341	79.021±0.285	72.102±0.329	60.002±0.326	50.210±0.347	79.342±0.282	71.512±0.346
ICT-META w/o	72.041±0.298	61.034±0.329	80.123±0.290	69.482±0.338	57.028±0.329	48.007±0.354	82.045±0.285	73.512±0.352

The data presented in this table was obtained through 4000 rounds of meta-testing using all pre-trained models. The ‘web‘ column refers to a combined dataset where all categories from ABC, Wiki, and SMH are merged for meta-testing. ICT-META* refers to the model trained using Randomly Initialized Embeddings. ICT-META refers to the model trained using the ELMES encoding mechanism. ICT-META* w/o and ICT-META w/o refers to the model trained only on DC Dataset (video + audio) without Google Home Dataset. † indicates methods requiring fine-tuning during meta-testing.

V. RESULTS

A. EVALUATION AND ANALYSIS OF ENCRYPTED TRAFFIC DATA ENCODER

1) SETTINGS

The YouTube dataset contains 30 video categories, and the YouTube Music dataset includes 10 audio categories from the DC Dataset. The model was trained for 30 epochs with a learning rate of 0.0001 and the Adam optimizer. The ReduceLROnPlateau scheduler dynamically adjusted the learning rate based on validation performance.

2) EVALUATION AND ANALYSIS

After 30 epochs, the model achieved a 75.50% testing accuracy, demonstrating its ability to extract critical features from encrypted network streams. This highlights the effectiveness of the Encrypted Traffic Data Encoder in encrypted traffic classification.

B. EVALUATION AND ANALYSIS OF ICT-META

1) SETTINGS

ICT-META was optimized using Adam with an initial learning rate of 0.0001, adjusted by a Custom Cosine scheduler over 40 epochs. The batch size was set to 75 for balanced efficiency. Key parameters are summarized in Table 1.

In the meta-testing phase, the model underwent 4,000 trials with 5-way classification, using either 1-shot or 5-shot support samples and 16 query samples per class. This setup tested the model's generalization ability.

2) EVALUATION AND ANALYSIS

The ICT-META model was evaluated on unseen web traffic data and compared against baseline models: MetaNIW, SNAIL, and GPICL. Results, presented in Table 2, show classification accuracy under 5-way 5-shot and 5-way 1-shot configurations.

3) ABLATION STUDY ON TRAINING DATASETS

To further investigate the effect of including multiple datasets during training, we conduct an ablation study using ICT-META and ICT-META* trained only on the DC Dataset (video, audio stream data) without the Google Home Dataset. As shown in Table II, removing the Google Home Dataset slightly decreases the model's performance on the web stream data across both 5-way 5-shot and 1-shot settings. This result indicates that incorporating additional data sources in the training phase contributes to better cross-domain generalization. Specifically, ICT-META w/o achieves 82.05% on web 5-shot, compared to 85.67% when trained with both DC and Google Home datasets.

Key Findings:

1) **Superior Performance of ICT-META**: The ICT-META model consistently outperforms the baseline models across all datasets and settings. In the 5-way 5-shot configuration, ICT-META achieves the highest accuracy of 73.91% on the “abc” dataset, compared to MetaNIW's 72.22%, and far surpasses SNAIL, which achieves only 20.24%. Additionally, on the “web” dataset, ICT-META achieves 85.67% and 75.90% in the 5-way 5-shot and 5-way 1-shot configurations, respectively, making it the best-performing model across all settings.
2) **Impact of ELMES Encoding**: When comparing ICT-META and ICT-META* (the version using randomly initialized embeddings), we observe that the use of the ELMES encoding mechanism significantly improves the performance. For example, on the “wiki” dataset in the 5-way 1-shot setting, ICT-META achieves 71.32%, while ICT-META* only achieves 74.45%. This confirms that ELMES encoding enhances the model's ability to extract meaningful features from encrypted traffic, contributing to better generalization across diverse datasets.
3) **Cross-Domain Adaptability**: One of the key strengths of ICT-META is its ability to perform well in cross-domain learning scenarios. On the combined “web” dataset, which includes categories from all three source datasets, ICT-META achieves an

accuracy of 85.67% in the 5-way 5-shot configuration. This demonstrates that ICT-META is capable of adapting to new, unseen traffic data and effectively generalizing across domains without the need for retraining or fine-tuning. In future work, we plan to further validate the robustness of ICT-META by evaluating multiple random partitions and training/testing splits across different service categories.

VI. CONCLUSION

This section summarizes the proposed In-Context Traffic Meta-learner(ICT-META) method for encrypted traffic classification. Traditional deep learning models rely heavily on large labeled datasets and require retraining or fine-tuning for new tasks, which can be time-consuming and resource-intensive. To address these issues, the ICT-META method combines few-shot learning with universal meta-learning, enabling the model to learn new traffic features directly from the context during inference without the need for fine-tuning, allowing it to quickly adapt to new types of encrypted traffic.

ICT-META uses non-causal sequence modeling, which helps the model adapt to new encrypted traffic types without retraining. Additionally, ICT-META utilizes a pre-trained residual network as a feature extractor and incorporates the ICT-META encoding mechanism, which enhances the model's feature extraction capabilities for encrypted traffic classification. Experimental results demonstrate that ICT-META outperforms traditional models, providing a faster and more general solution for encrypted traffic classification, capable of quickly learning new features and accurately classifying encrypted traffic of various types. Furthermore, our ablation study further demonstrates the importance of leveraging multiple datasets during training. Models trained only on the DC Dataset (video, audio stream data) exhibit slightly lower cross-domain performance, confirming that including diverse data sources improves the model's generalization capability.

Although ICT-META shows promising performance, especially in cross-domain encrypted traffic classification scenarios, several limitations should be acknowledged. First, ICT-META operates on fixed-length traffic feature representations, and its robustness under higher-dimensional traffic features has not yet been fully evaluated. Second, the Transformer-based in-context modeling introduces additional computational overhead, which may pose challenges in resource-constrained or large-scale deployment environments. Third, while ICT-META adapts well to new traffic types, its current accuracy on certain categories such as SMH may not meet the latency and reliability requirements of real-time security systems. Finally, in some datasets with relatively structured traffic patterns like the Wiki dataset, simpler embedding strategies may perform competitively in low-shot scenarios, highlighting potential limitations of the ELMES encoding in specific contexts. Addressing these limitations, such as enhancing class-wise consistency, supporting high-dimensional traffic, and developing lightweight or incremental inference strategies, constitutes an important direction for future work.

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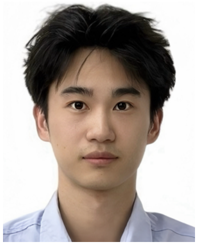

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